

The ShoppyGlobe API consists of 3 demo products with unique Id, product name, price and quantities in stock. The following screenshots will show testing done in the ThunderClient and MongoDB.

The screenshots include:

1. GET all products and single product

2. Register user

3. Login user

4. Authorization

5. POST add to cart

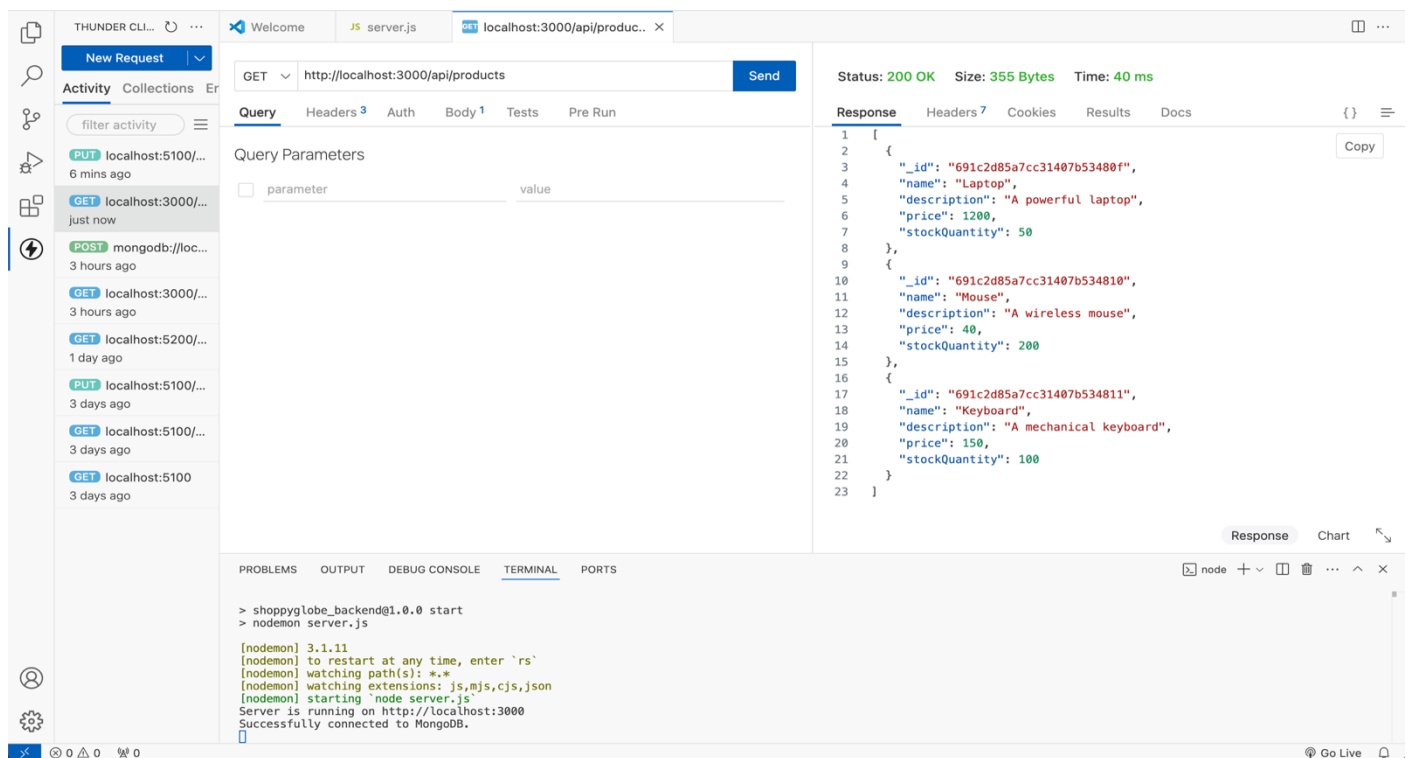
6. Get User's cart

7. PUT update cart quantity

8. DELETE remove from cart

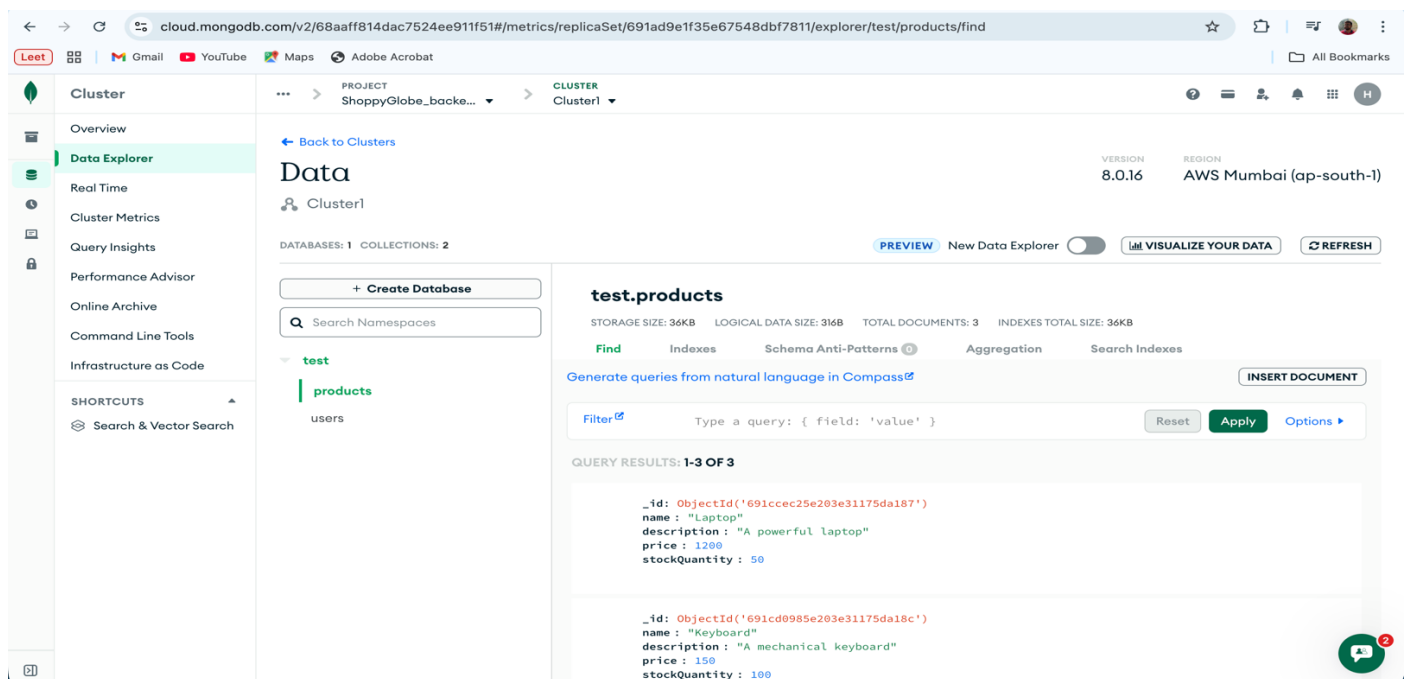
9. Error handling

1.GET all products and single product



The screenshot shows the ThunderClient interface. A GET request is sent to `http://localhost:3000/api/products`. The response is a JSON array of three products:

```
1 {
2   {
3     "_id": "691c2d85a7cc31407b53480f",
4     "name": "Laptop",
5     "description": "A powerful laptop",
6     "price": 1200,
7     "stockQuantity": 50
8   },
9   {
10    "_id": "691c2d85a7cc31407b534810",
11    "name": "Mouse",
12    "description": "A wireless mouse",
13    "price": 40,
14    "stockQuantity": 200
15  },
16  {
17    "_id": "691c2d85a7cc31407b534811",
18    "name": "Keyboard",
19    "description": "A mechanical keyboard",
20    "price": 150,
21    "stockQuantity": 100
22  }
23 }
```



The screenshot shows the MongoDB Cloud interface. The `test.products` collection is selected. The collection contains three documents:

```
_id: ObjectId('691c2d85a7cc31407b53480f')
name: "Laptop"
description: "A powerful laptop"
price: 1200
stockQuantity: 50

_id: ObjectId('691c2d85a7cc31407b534810')
name: "Mouse"
description: "A wireless mouse"
price: 40
stockQuantity: 200

_id: ObjectId('691c2d85a7cc31407b534811')
name: "Keyboard"
description: "A mechanical keyboard"
price: 150
stockQuantity: 100
```

THUNDER CLI... Welcome JS server.js localhost:3000/api/produ...

New Request

Activity Collections Er

filter activity

GET localhost:3000/... just now

PUT localhost:5100/... 36 mins ago

POST mongodb://loc... 3 hours ago

GET localhost:3000/... 4 hours ago

GET localhost:5200/... 1 day ago

PUT localhost:5100/... 3 days ago

GET localhost:5100/... 3 days ago

GET localhost:5100 3 days ago

GET http://localhost:3000/api/products/691c2d85a7cc31407b53480f Send

Query Headers 3 Auth Body 1 Tests Pre Run

Query Parameters

parameter value

Status: 200 OK Size: 116 Bytes Time: 8 ms

Response Headers 7 Cookies Results Docs

```
1 {
2   "_id": "691c2d85a7cc31407b53480f",
3   "name": "Laptop",
4   "description": "A powerful laptop",
5   "price": 1200,
6   "stockQuantity": 50
7 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> shoppyglobe_backend@1.0.0 start
> nodemon server.js

[nodemon] 3.1.11
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

Go Live

2. Register user

THUNDER CLI... Welcome JS server.js localhost:3000/api/produ...

New Request

Activity Collections Er

filter activity

POST localhost:300... just now

PUT localhost:5100/... 36 mins ago

POST mongodb://loc... 3 hours ago

GET localhost:3000/... 4 hours ago

GET localhost:5200/... 1 day ago

PUT localhost:5100/... 3 days ago

GET localhost:5100/... 3 days ago

GET localhost:5100 3 days ago

POST http://localhost:3000/api/auth/register Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
1 {
2   "username": "Hrudendu",
3   "password": "password"
4 }
```

Status: 201 Created Size: 43 Bytes Time: 94 ms

Response Headers 7 Cookies Results Docs

```
1 {
2   "message": "User registered successfully!"
3 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> shoppyglobe_backend@1.0.0 start
> nodemon server.js

[nodemon] 3.1.11
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

Go Live

3.Login user

THUNDER CLI... Welcome JS server.js localhost:3000/api/produ...

New Request

Activity Collections Er

filter activity

POST localhost:300... just now

PUT localhost:5100/... 36 mins ago

POST mongodb://loc... 3 hours ago

GET localhost:3000/... 4 hours ago

GET localhost:5200/... 1 day ago

PUT localhost:5100/... 3 days ago

GET localhost:5100/... 3 days ago

GET localhost:5100... 3 days ago

POST http://localhost:3000/api/cart/ Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content

```
1 {
2   "productId": "691c2d85a7cc31407b53480f",
3   "quantity": 5
4 }
```

Status: 200 OK Size: 132 Bytes Time: 68 ms

Response Headers 7 Cookies Results Docs

```
1 {
2   "message": "Product added to cart.",
3   "cart": {
4     "productId": "691c2d85a7cc31407b53480f",
5     "quantity": 5,
6     "_id": "691ca61317568c57adf081f0"
7   }
8 }
9
10 }
```

Problems Output Debug Console Terminal Ports

```
> shoppyglobe_backend@1.0.0 start
> nodemon server.js

[nodemon] 3.1.11
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

6. GET user's cart

EXPLORER SHOPPYGLOBE_BACKEND

- controllers
- JS auth.controller.js
- JS cart.controller.js
- JS product.controller.js
- middleware
- JS auth.middleware.js
- models
- JS product.model.js
- JS user.model.js
- node_modules
- routes
- JS auth.routes.js
- JS cart.routes.js
- JS product.routes.js
- { } package-lock.json
- { } package.json
- JS server.js

GET http://localhost:3000/api/cart/ Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content

```
1 {
2   "productId": "691c2d85a7cc31407b53480f",
3   "quantity": 5
4 }
```

Status: 200 OK Size: 178 Bytes Time: 11 ms

Response Headers 7 Cookies Results Docs

```
1 {
2   "productId": {
3     "_id": "691c2d85a7cc31407b53480f",
4     "name": "Laptop",
5     "description": "A powerful laptop",
6     "price": 1200,
7     "stockQuantity": 50
8   },
9   "quantity": 5,
10   "_id": "691ca61317568c57adf081f0"
11 }
12
13 }
```

Problems Output Debug Console Terminal Ports

```
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

7. PUT update cart quantity

Visual Studio Code interface showing a REST client request and response.

Request:

- Method: PUT
- URL: http://localhost:3000/api/cart/691c2d85a7cc31407b53480f
- Body (JSON):

```
{  "quantity": 3}
```

Response:

- Status: 200 OK
- Size: 88 Bytes
- Time: 17 ms
- Response Body (JSON):

```
{  "productId": "691c2d85a7cc31407b53480f",  "quantity": 3,  "_id": "691ca61317568c57adf081f0"}
```

Terminal Output:

```
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting 'node server.js'
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

MongoDB Compass interface showing the 'users' collection.

Documents:

```
{  "_id": {},  "username": "Hrudendu",  "password": "$2b$08$rY6txPi2zUd1QerxJZPTy.nuaxE8zh7KbEjWvI.mZb8LgYJHDPXZ6",  "cart": [    {      "productId": {      },      "soid": "691c2d85a7cc31407b53480f",      "quantity": 3,      "_id": {      },      "soid": "691ca61317568c57adf081f0"    }  ],  "__v": 1}
```

8. DELETE remove from cart

EXPLORER

- SHOPPYGLOBE_BACKEND
 - controllers
 - auth.controller.js
 - cart.controller.js
 - product.controller.js
 - middleware
 - auth.middleware.js
 - models
 - product.model.js
 - user.model.js
 - node_modules
 - routes
 - auth.routes.js
 - cart.routes.js
 - product.routes.js
 - package-lock.json
 - package.json
 - server.js

DELETE http://localhost:3000/api/cart/691c2d85a7cc31407b53480f Send

Query Headers 3 Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content

```
1
```

Status: 200 OK Size: 2 Bytes Time: 33 ms

Response Headers 7 Cookies Results Docs

```
1 []
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

9. Error Handling

9.1. Authentication error- one cannot access cart items if user login token is not provided

EXPLORER

- SHOPPYGLOBE_BACKEND
 - controllers
 - auth.controller.js
 - cart.controller.js
 - product.controller.js
 - middleware
 - auth.middleware.js
 - models
 - product.model.js
 - user.model.js
 - node_modules
 - routes
 - auth.routes.js
 - cart.routes.js
 - product.routes.js
 - package-lock.json
 - package.json
 - server.js

GET http://localhost:3000/api/cart Send

Query Headers 2 Auth Body Tests Pre Run

HTTP Headers

☒ Accept */*

☒ User-Agent Thunder Client (https://www.thunderclient.com)

☐ Authorization JWT eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1bm90b3R5IjoiYWRhcm91IiwiaWF0IjoxNjU0MjU0MjU0.eyJ1bm90b3R5IjoiYWRhcm91IiwiaWF0IjoxNjU0MjU0MjU0

☐ header value

Status: 403 Forbidden Size: 32 Bytes Time: 23 ms

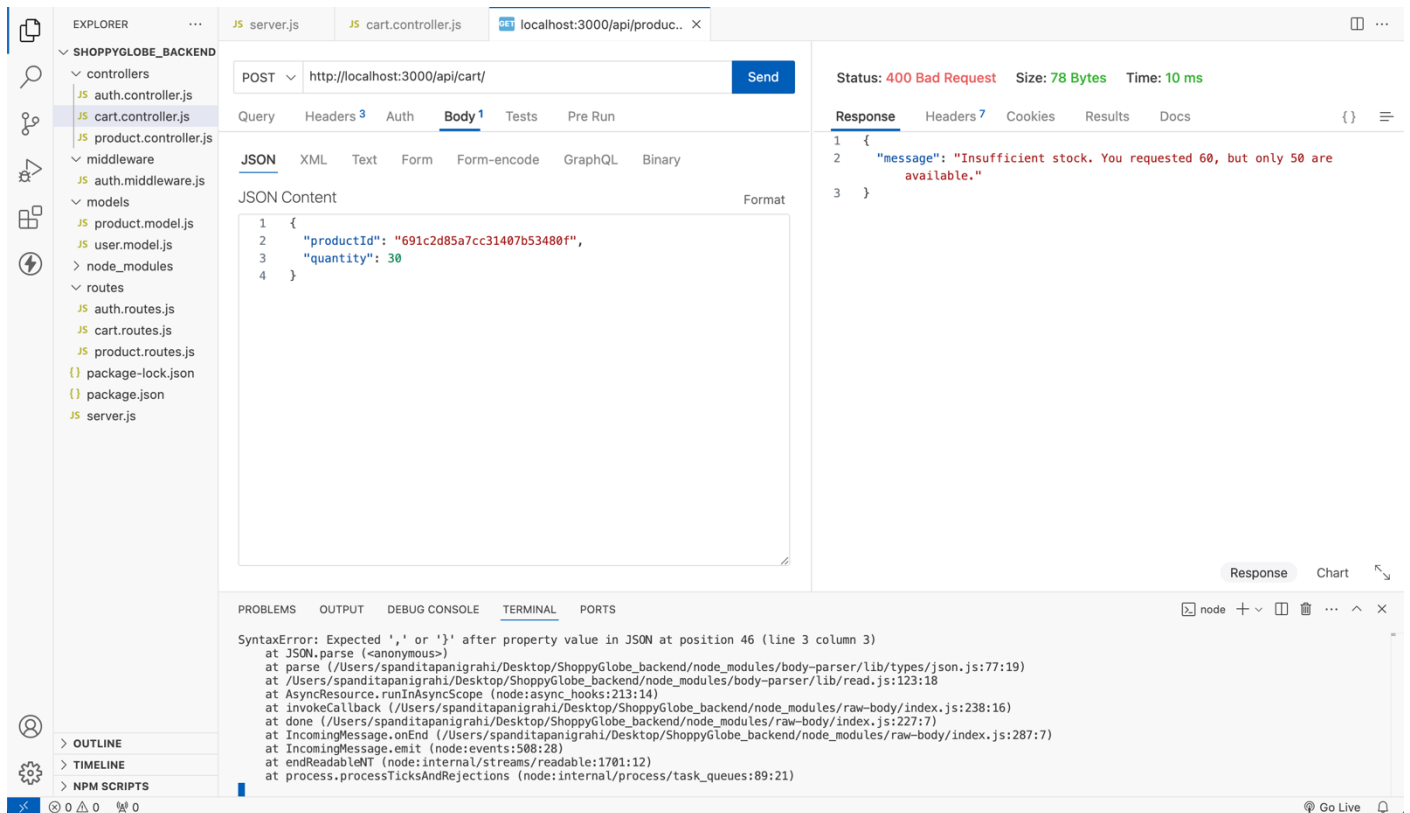
Response Headers 7 Cookies Results Docs

```
1 {
2   "message": "No token provided!"
3 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server is running on http://localhost:3000
Successfully connected to MongoDB.
```

9.2. Insufficient Stock- final cart item of a product should not exceed stock quantity of that product



The screenshot shows the VS Code interface with the REST client open. The request is a POST to `http://localhost:3000/api/cart/` with a JSON body:

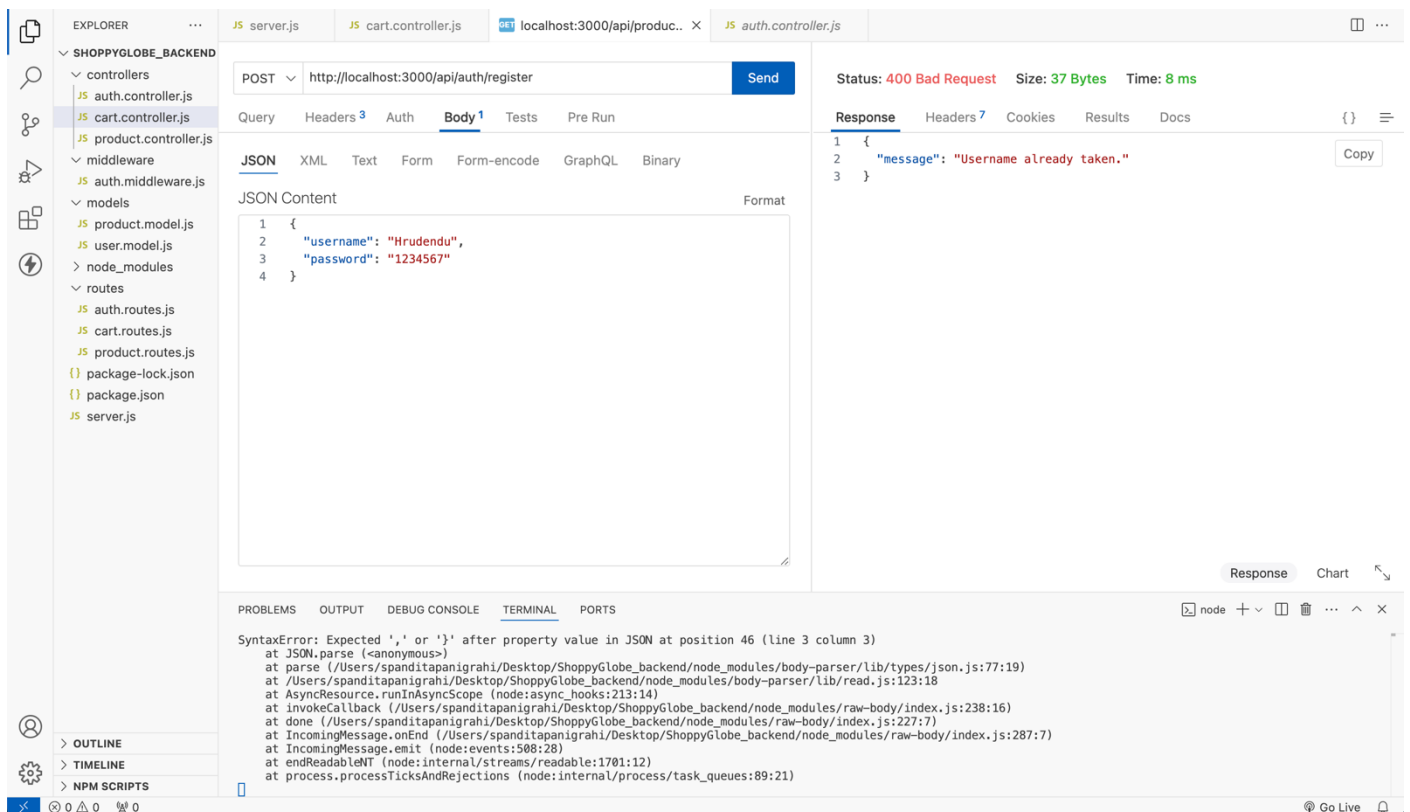
```
1 {
2   "productId": "691c2d85a7cc31407b53480f",
3   "quantity": 30
4 }
```

The response is a 400 Bad Request with a message: "Insufficient stock. You requested 60, but only 50 are available."

The terminal shows a syntax error in the JSON body:

```
SyntaxError: Expected ',' or '}' after property value in JSON at position 46 (line 3 column 3)
at JSON.parse (<anonymous>)
at parse (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/body-parser/lib/types/json.js:77:19)
at /Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/body-parser/lib/read.js:123:18
at AsyncResource.runInAsyncScope (node:async_hooks:213:14)
at invokeCallback (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:238:16)
at done (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:227:7)
at IncomingMessage.onEnd (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:287:7)
at IncomingMessage.emit (node:events:508:28)
at endReadableNT (node:internal/stream_base/large_objects:1701:12)
at process.processTicksAndRejections (node:internal/process/task_queues:89:21)
```

9.3 Username already registered



The screenshot shows the VS Code interface with the REST client open. The request is a POST to `http://localhost:3000/api/auth/register` with a JSON body:

```
1 {
2   "username": "Hrudendu",
3   "password": "1234567"
4 }
```

The response is a 400 Bad Request with a message: "Username already taken."

The terminal shows a syntax error in the JSON body:

```
SyntaxError: Expected ',' or '}' after property value in JSON at position 46 (line 3 column 3)
at JSON.parse (<anonymous>)
at parse (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/body-parser/lib/types/json.js:77:19)
at /Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/body-parser/lib/read.js:123:18
at AsyncResource.runInAsyncScope (node:async_hooks:213:14)
at invokeCallback (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:238:16)
at done (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:227:7)
at IncomingMessage.onEnd (/Users/spanditapanigrahi/Desktop/ShoppyGlobe_backend/node_modules/raw-body/index.js:287:7)
at IncomingMessage.emit (node:events:508:28)
at endReadableNT (node:internal/stream_base/large_objects:1701:12)
at process.processTicksAndRejections (node:internal/process/task_queues:89:21)
```